

Joy Zhang

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EDUCATION

Cornell University, College of Engineering

Master of Engineering, Computer Science, GPA 3.9

May 2021

Bachelor of Science, Computer Science Major, Game Design Minor, GPA 3.5

Dec 2020

PROFESSIONAL EXPERIENCE

Sandtable

May 2025 - Present

Game Engineer

- Develop game loop, gameplay mechanics, UI, and shaders for a multiplayer RTS Unity game

Hometopia Inc.

Senior Game Developer

Dec 2024 - Feb 2025

Junior Game Developer

Jul 2024 - Dec 2024

- Contributed to the development of a multiplayer house-building PC sandbox game in Unity by writing clean, well-organized, and easily maintainable C# code
- Took ownership of UI development, collaborated with designers to give game a cohesive look and feel
- Lead and mentored a small team of junior developers, delegated tasks, and performed code reviews
- Developed an audio system to centralize control of music, ambience, and sound effects
- Created a system for the first-time user experience and tutorial
- Improved multiplayer features, including friend list, parties, chat, and save file transfer and sharing
- Investigated and fixed critical gameplay, multiplayer, and UI bugs, and developed new features
- Took responsibility for developing demo of a separate virtual pet mobile game and setting up structure of game by implementing a state machine for main game loop and connecting different game systems

Northrop Grumman

ODIN, Game Developer

Jan 2022 - Jun 2024

- Developed Omniverse apps in Python with Universal Scene Description to build a virtual warfighting environment by controlling drones and sensors, streamlining data generation, and visualizing data
- Built physically accurate flight simulators for Volansi and Spirit drones with Omniverse's integrated PhysX physics engine and performed manual 3D vector math calculations
- Integrated Omniverse simulation with external AI code and live drones running in real-time
- Created a data generation pipeline for Omniverse to train AI/ML models by developing a suite of tools in Python to create, control, and randomize scenes, assets, and perceptual variables

COAGEN, Software Engineer

Aug 2021 - Dec 2021

- Built demo in Java for a long short-term memory (LSTM) predicting adhoc interruptions to schedule

PROJECTS

Game Design Initiative at Cornell

Jan 2019 - May 2021

- Developed gameplay mechanics, physics, and programmatic animations with the Cornell University Games Library in C++ and libGDX in Java as a gameplay programmer to make games in teams of 7
- Accepted to BostonFIG Fest with computer game *Starstruck* (2019) and mobile game *Spectacle!* (2020)
- Guided students through entire process of designing and creating digital games as a Teaching Assistant

Global Summit

Jun 2020 - Aug 2020

- Created Pygame simulation for an unmanned wildfire identification and suppression system

Cornell Big Red Hacks

Sep 2019

- Awarded Best Design Hack for *One Call Away*, a short story platformer created in Unity in 36 hours

SKILLS

Competencies: Agile, Algorithms, Data structures, Game development, Gameplay programming, Object-oriented programming, Physics simulation, UI programming

Programming Languages: C, C#, C++, Java, MATLAB, OCaml, Perl, PHP, Python

Specialties: Blender, Box2D, Construct3, libGDX, Nvidia Omniverse, PhysX, Pygame, Unity

Other Software: Adobe (Character Animator, Illustrator), ClickUp, CSS, Docker, Eclipse, Git, HTML, IntelliJ IDEA, Jira, LaTeX, Linux, Maven, Plastic SCM, Robot Framework, Visual Studio, Visual Studio Code